Performance Analysis of 3D Palmprint Human Biometric Recognition System using Hierarchical Spatial Attention Network

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***Abstract*—This paper presents a hybrid 3D palmprint biometric recognition framework that integrates pseudo-3D reconstruction with attention-based deep learning to enhance identification performance. The proposed approach utilizes the Frankot–Chellappa algorithm to reconstruct depth-enhanced palm surfaces from single-view 2D images, improving robustness to illumination variations and positional inconsistencies. To evaluate feature extraction strategies, a comparative analysis is conducted using PCANet, MobileNetV3, and a Hierarchical Spatial Attention Network (HSANet). PCANet provides a computationally efficient baseline, while MobileNetV3 offers a balance between accuracy and computational efficiency. In contrast, HSANet achieves superior performance by leveraging spatial attention mechanisms to focus on discriminative palm regions. Experimental results on the IIT Delhi palmprint dataset demonstrate that the proposed framework significantly improves recognition accuracy and reduces error rates, achieving near-perfect classification under controlled conditions. Performance is evaluated using standard biometric metrics, including accuracy, Equal Error Rate (EER), Receiver Operating Characteristic (ROC), and Cumulative Match Characteristic (CMC) curves. Furthermore, the complete recognition pipeline is implemented and deployed on an NVIDIA Jetson Nano platform. The system demonstrates efficient resource utilization, stable inference performance, and practical feasibility for real-time biometric authentication in resource-constrained edge environments. Although the results are promising, further validation on unconstrained datasets is required to ensure robustness under real-world conditions.**

***Index Terms*—2D Palmprint images, 3D reconstruction, Feature extraction, Classification, False acceptance rate(FAR), Equal error rate(EER).**

I. Introduction

Biometric authentication has emerged as a reliable solution for secure identity verification due to its ability to utilize unique physiological characteristics such as palmprints, fingerprints, and iris patterns [1], [2]. Among these modalities, palmprint recognition has gained significant attention owing to its rich set of discriminative features, including principal lines, wrinkles, and ridge structures.

Traditional palmprint recognition systems primarily rely on two-dimensional (2D) images [14], [15]. However, such approaches are sensitive to variations in illumination, hand positioning, and surface deformation, which can degrade recognition performance. To address these limitations, recent research has explored three-dimensional (3D) biometric representations, as they provide additional depth and structural information [5], [14].

Despite their advantages, most existing 3D biometric systems require specialized acquisition hardware or multi-view imaging setups, making them computationally expensive and unsuitable for real-time deployment on resource-constrained devices [5]. Furthermore, conventional feature extraction techniques often fail to effectively capture spatially localized discriminative patterns.

To overcome these challenges, this work proposes a hybrid framework that combines pseudo-3D reconstruction using the Frankot–Chellappa algorithm with advanced feature extraction using PCANet and a Hierarchical Spatial Attention Network (HSANet). The proposed approach enables efficient depth estimation from single-view images while leveraging attention mechanisms to focus on discriminative palm regions.

The system is further optimized for deployment on embedded hardware (Jetson Nano), demonstrating its feasibility for real-time biometric authentication applications.

A lightweight deep learning architecture (MobileNetV3) is also incorporated as a baseline to evaluate the effectiveness of learned feature representations in biometric recognition. Its efficient design enables a balance between recognition performance and computational complexity, making it suitable for comparison with both PCANet and attention-based models such as HSANet.

Novelty And Contributions

In contrast to existing palmprint identification systems that only use 2D texture features or computationally expensive real 3D acquisition systems, this work proposes a hybrid pseudo-3D framework that blends gradient-based surface reconstruction with attention-guided deep learning.

A technique that uses Frankot–Chellappa to rebuild pseudo-3D palmprints from single-view 2D pictures without the need for multi-view gear. Learning spatially adaptive characteristics on rebuilt surfaces using the Hierarchical Spatial Attention Network (HSANet).A comprehensive comparison of PCANet, MobileNetV3, and HSANet is performed under identical experimental conditions to analyze the trade-off between computational efficiency and recognition performance across different feature extraction paradigms.

The full pipeline is deployed on resource-constrained edge hardware (Jetson Nano) to demonstrate real-time capability. This combination of reconstruction, attention, and edge deployment is not jointly examined in the published studies.Unlike existing approaches, this work jointly integrates pseudo-3D reconstruction, attention-based feature learning, and real-time edge deployment within a unified framework.

II. RELATED WORK

Recent research in biometric recognition has focused on improving accuracy and robustness through multimodal systems, advanced feature extraction techniques, and three-dimensional (3D) representations. Early palmprint recognition approaches relied on handcrafted features, which demonstrated limited generalization under varying environmental conditions.

Multimodal biometric systems have been widely explored to enhance recognition performance. For instance, multimodal biometric systems combining palmprint and finger knuckle features have demonstrated improved recognition accuracy through feature fusion [1].

Contactless biometric systems have also gained attention due to their improved usability and hygiene. Contactless biometric systems have been proposed to eliminate physical interaction with sensors; however, they remain sensitive to illumination and pose variations [4].

The use of three-dimensional biometric representations has further improved system robustness by incorporating depth and structural information. Studies have shown that 3D biometrics can significantly enhance recognition accuracy compared to traditional 2D approaches [14], [15]. However, many existing 3D systems rely on specialized hardware or multi-view acquisition techniques, making them computationally expensive and less suitable for real-time embedded applications [5].

Deep learning techniques have significantly advanced feature extraction in biometric systems by enabling automatic learning of discriminative representations. However, conventional convolutional neural networks often treat all spatial regions equally, which may reduce effectiveness when only specific regions contain relevant information.

To address this limitation, attention-based models have been proposed to selectively focus on informative regions. These models enhance feature representation by emphasizing discriminative areas while suppressing irrelevant information, leading to improved recognition performance.

Despite these advancements, there remains a lack of systematic comparison between lightweight models and attention-based deep learning approaches within a unified framework. Additionally, many existing methods either require high computational resources or lack suitability for deployment on embedded systems.

In contrast, this work proposes a hybrid framework that combines pseudo-3D reconstruction with attention-based feature learning. Furthermore, a comprehensive comparison between PCANet and HSANet is performed under identical experimental conditions, providing insights into the trade-off between computational efficiency and recognition performance in 3D palmprint biometrics.

III. PROPOSED METHOD

The proposed palmprint recognition system consists of four main stages: preprocessing, pseudo-3D reconstruction, feature extraction, and classification. The input is a 2D palmprint dataset obtained from the IIT Delhi database, containing high-resolution images with rich palm features.

Initially, the images are preprocessed to reduce noise and normalize illumination variations. A consistent Region of Interest (ROI) is then extracted to isolate the central palm region. The Frankot–Chellappa algorithm is applied to the ROI to reconstruct a pseudo-3D palm surface using gradient-based estimation from single-view images. This enhances robustness against illumination and positional variations.

Following reconstruction, feature extraction is performed using both PCANet and HSANet models. PCANet provides a computationally efficient baseline, while HSANet employs attention mechanisms to focus on discriminative palm regions. Finally, a supervised classification model is used to perform identity recognition based on the extracted features.

The overall system is designed to achieve accurate and hardware-efficient biometric authentication.

A. 2D multiple images acquisition

Single-view photos taken under controlled circumstances make up the IIT Delhi palmprint dataset. In this work, gradient-based estimation from 2D pictures is used for 3D reconstruction instead of actual multi-view acquisition.

B. 3D Reconstruction

The 3D reconstruction is performed using gradient-based estimation from single-view 2D palmprint images. 3D reconstruction produces an accurate 3D model of the palmprint.

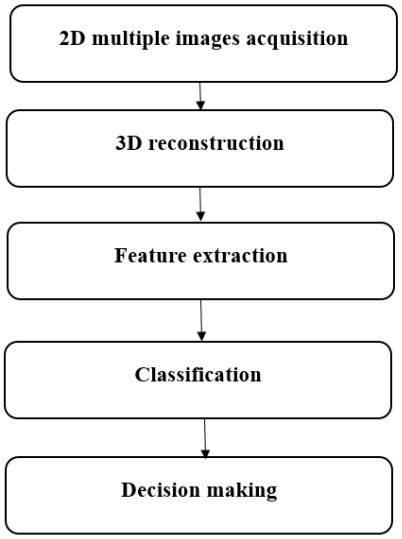

Fig. 1. Proposed Block Diagram

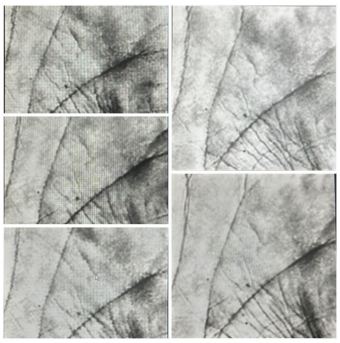

Fig. 2. Sample of 2D multiple images

A sample of 2D multiple input palmprint photos is used for 3D reconstruction using Frankot Chellappa (FC). 3D palmprint images offer depth estimation and surface detail retention. The Frankot Chellappa method (FC) ensures accurate line, wrinkle, and ridge reconstruction for biometric authentication analysis. Since the usage of multiple input photos enhances depth perception and geometric consistency, the rebuilt 3D model generated using MATLAB R2024B is utilized for additional processing and performance evaluation.

The Frankot–Chellappa algorithm is widely used for reconstructing a three-dimensional surface from gradient information. The process begins with gradient fields $p(x, y)$ and $q(x, y)$, which represent the surface gradients along the x- and y-directions, respectively, obtained from multiple two-dimensional palmprint images. These gradients are defined as:

$$p(x, y) = \frac{\partial z(x, y)}{\partial x}, \quad q(x, y) = \frac{\partial z(x, y)}{\partial y} \tag{1}$$

where $z(x, y)$ denotes the surface height. The relationship between the gradients and the surface is formulated as a Partial Differential Equation (PDE):

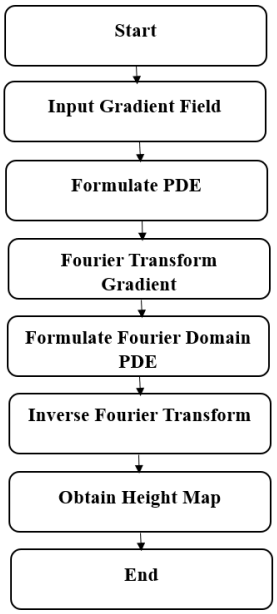

Fig. 3. Frankot Chellappa Flowchart for Generating Height Map from Gradient Fields

$$\frac{\partial z}{\partial x} = p(x, y), \quad \frac{\partial z}{\partial y} = q(x, y) \tag{2}$$

To solve this efficiently, the gradients are transformed into the frequency domain using the Fourier transform:

$$P(u, v) = \mathcal{F}\{p(x, y)\}, \quad Q(u, v) = \mathcal{F}\{q(x, y)\} \tag{3}$$

In the frequency domain, differentiation becomes multiplication, leading to:

$$j2\pi u\, Z(u, v) = P(u, v), \quad j2\pi v\, Z(u, v) = Q(u, v) \tag{4}$$

where $Z(u, v)$ is the Fourier transform of the surface $z(x, y)$. The optimal solution that enforces integrability is given by:

$$Z(u, v) = \frac{j2\pi u\, P(u, v) + j2\pi v\, Q(u, v)}{(2\pi u)^2 + (2\pi v)^2} \tag{5}$$

for $(u, v) \neq (0, 0)$, and $Z(0, 0) = 0$ to avoid singularity. Finally, the reconstructed surface is obtained by applying the inverse Fourier transform:

$$z(x, y) = \mathcal{F}^{-1}\{Z(u, v)\} \tag{6}$$

This process generates a consistent height map in which each pixel represents the relative depth of the palm surface. The reconstructed three-dimensional surface preserves important structural features such as ridges, wrinkles, and principal lines, which are essential for subsequent feature extraction and biometric recognition. The 3D reconstructed image using the Frankot–Chellappa algorithm is shown in Fig. 4..

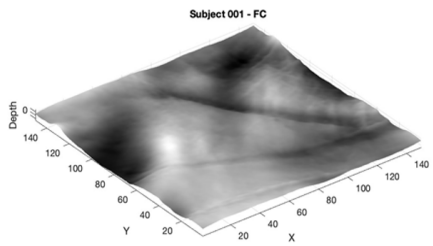

Fig. 4. 3D Reconstruction using Frankot Chellappa algorithm

IV. Feature Extraction and Classification

A. Feature Extraction

Although it impacts how successfully a palm's distinctive patterns are captured for identification, feature extraction is an essential stage in palmprint biometric recognition. In this work, significant features are extracted from reconstructed 3D palmprint data using two deep learning-based feature extraction models, PCANet and HSANet. PCANet is a straightforward and effective model that uses cascaded Principal Component Analysis (PCA) filters to learn texture features, making it appropriate for systems with constrained processing power. However, its capacity to record intricate and extremely distinctive 3D palm patterns is constrained due to its reliance on linear feature learning. HSANet, on the other hand, employs an attention-based learning strategy that minimizes the impact of background noise while automatically focusing on the most significant areas of the palm, such as main lines, ridges, and depth changes. HSANet outperforms PCANet in recognition performance, particularly under difficult biometric settings, thanks to its concentrated feature learning, which produces more robust and discriminative representations.

B. Classification

To identify the person, a classifier receives the palmprint features after feature extraction. In this work, a supervised classification model that differentiates between real and fake data is fed features from both PCANet and HSANet. In order to allocate each test sample to the most suitable class, the classifier establishes decision boundaries after learning how the characteristics are distributed. There is increased overlap between classes as a result of PCANet features' lower discriminative power, which raises the number of misclassifications. On the other hand, the classifier can make more accurate decisions with lower error rates because the features collected using HSANet are more compact and clearly differentiated.This demonstrates unequivocally that robust and targeted feature extraction is necessary for trustworthy classification, and the attention-based features that HSANet learns result in more accurate and consistent palmprint identification.

A supervised classification approach is used to perform final classification, where the extracted feature vectors are mapped to their corresponding classes. The classifier learns discriminative decision boundaries based on the feature representations obtained from PCANet and HSANet. The model is trained using an appropriate loss function to minimize classification error and improve recognition performance.

C. HSANet Architecture

The Hierarchical Spatial Attention Network (HSANet) is designed to enhance feature representation by integrating multi-level spatial attention with hierarchical feature learning. Given an input image $X \in \mathbb{R}^{H \times W \times C}$, a convolutional backbone (e.g., ResNet) extracts feature maps at different stages $\{F_1, F_2, \ldots, F_n\}$, where each $F_i \in \mathbb{R}^{H_i \times W_i \times C_i}$ encodes progressively higher-level semantics.

For each feature map, a spatial attention mechanism is applied to emphasize informative regions. The attention map is computed as:

$$A_i = \sigma\big(f^{k \times k}(F_i)\big) \tag{7}$$

where $f^{k \times k}(\cdot)$ denotes a convolution operation and σ is the sigmoid activation. The refined feature is obtained via element-wise multiplication:

$$F_i' = F_i \odot A_i \tag{8}$$

This operation suppresses irrelevant background while enhancing salient regions (e.g.,salient palm regions). To preserve original information and stabilize training, residual connections are incorporated:

$$\hat{F}_i = F_i + F_i' \tag{9}$$

HSANet further exploits a hierarchical structure, where attention-refined features from multiple scales are fused to capture both local and global context. The fusion process is typically expressed as:

$$F_{\text{fusion}} = \text{Concat}(\hat{F}_1, \hat{F}_2, \ldots, \hat{F}_n) \tag{10}$$

or alternatively via summation. This multi-scale aggregation enables the network to detect fine-grained patterns alongside broader structural information. The fused feature map is then passed through global average pooling (GAP) to obtain a compact descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F_{\text{fusion}}(i, j, c) \tag{11}$$

which is subsequently fed into a fully connected layer for classification:

$$y = \text{Softmax}(Wz + b) \tag{12}$$

Overall, HSANet improves discriminative capability by hierarchically refining spatial features, enabling robust performance in tasks such as biometric recognition tasks, where capturing subtle texture variations and localized anomalies is

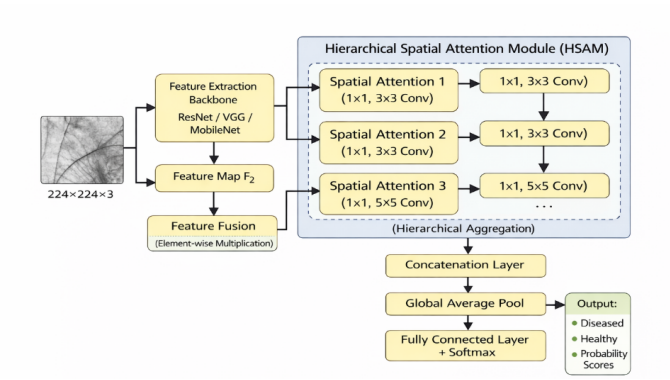

Fig. 5. HSANet Architecture

critical. The HSANet architecture is implemented using a convolutional backbone consisting of multiple convolutional layers with increasing channel depth (e.g., 64, 128, 256). Spatial attention modules are inserted after each stage to refine feature maps. The final feature representation is obtained through global average pooling followed by a fully connected layer for classification.

D. MobileNetV3 as a lightweight Baseline Model

In addition to PCANet and HSANet, MobileNetV3 is employed as a lightweight deep learning baseline model to evaluate feature extraction and classification performance. MobileNetV3 is designed for efficient computation on resource-constrained devices by utilizing depthwise separable convolutions and neural architecture search-based optimization.

The architecture incorporates squeeze-and-excitation (SE) modules and hard-swish activation functions to enhance feature representation while maintaining low computational complexity. Compared to traditional convolutional networks, MobileNetV3 significantly reduces the number of parameters and floating-point operations (FLOPs), making it suitable for real-time embedded applications.

In this work, MobileNetV3 processes the reconstructed 3D palmprint images and extracts discriminative feature representations, which are then used for classification. This model serves as an intermediate benchmark between lightweight handcrafted approaches such as PCANet and attention-based deep learning models such as HSANet.

V. Dataset and Experimental Setup

A. Dataset Description and Experimental Protocol

Details of the dataset:

There are about 230 participants in the IIT Delhi palmprint dataset, and each subject has six samples, for a total of 1380 grayscale photos.

Split Training and Testing: The dataset is divided into 70% for training and 30% for testing. A subject-disjoint protocol is followed to ensure no overlap between training and testing samples.

To ensure robustness, cross-validation is performed using a subject-disjoint training and testing split, and the results are averaged across multiple folds.

Important Information: The dataset is 2D by nature. In this work, gradient estimation is used instead of genuine multi-view acquisition for pseudo-3D reconstruction. When interpreting performance results, this distinction is crucial.

B. Preprocessing

All images were resized to 224×224 pixels and normalized to the range [0,1]. ROI extraction was performed using fixed coordinate cropping centered on the palm region. Gaussian filtering was applied to reduce noise, followed by histogram equalization to enhance contrast.

C. Evaluation Metrics (Detailed Description)

Accuracy is defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}} \tag{13}$$

The Equal Error Rate (EER) is obtained at the operating point where:

$$\text{FAR} = \text{FRR} \tag{14}$$

Receiver Operating Characteristic (ROC) curves are generated by varying the classification threshold and plotting the True Positive Rate (TPR) against the False Positive Rate (FPR), where:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN} \tag{15}$$

Cumulative Match Characteristic (CMC) curves are computed by ranking similarity scores and measuring the identification probability at rank-N.

By changing the classification threshold from 0 to 1 and calculating the True Positive Rate (TPR) and False Positive Rate (FPR) for each threshold value, ROC curves are produced. At the operating point where the False Acceptance Rate (FAR) and the False Rejection Rate (FRR) are equal, the Equal Error Rate (EER) is calculated.

Ranking similarity scores between probing and gallery samples creates CMC curves. Higher ranks show the likelihood of proper identification within the top-N matches, whereas Rank-1 accuracy is the probability that the correct identity emerges as the top match.

D. Training Configuration

The HSANet model was trained using a stochastic gradient-based optimizer with an appropriate learning rate. The batch size was set to 32, and the model was trained for 50 epochs. Categorical cross-entropy loss was used as the optimization objective.All evaluation curves were generated using MATLAB-based evaluation scripts derived from classifier output scores. All simulations and model implementations were carried out using MATLAB R2024B. The Deep Learning Toolbox was used for training the HSANet model, while image preprocessing and 3D reconstruction were implemented

using MATLAB's image processing and numerical computing functions.

The training process was performed using stochastic gradient descent with momentum (SGDM), with a batch size of 32 and 50 epochs. The implementation ensures reproducibility and consistency across all experiments.

VI. Simulation results

The IIT Delhi palmprint dataset consists of approximately 230 subjects, with 6 samples per subject, resulting in a total of 1380 grayscale images. The dataset is split into 70% for training and 30% for testing. All experiments are conducted using MATLAB-based implementation, ensuring consistent preprocessing, feature extraction, and evaluation across all models

The performance is evaluated using standard biometric metrics including accuracy, Equal Error Rate (EER), Receiver Operating Characteristic (ROC), and Cumulative Match Characteristic (CMC) curves.

Accuracy comparison between PCANet and HSANet is shown in Fig.5

TABLE I
Comparison of model accuracy, computation time and EER for different algorithms in simulation

Algorithms	Accuracy in percentage	Computation time	Equal error rate
PCANet	48.91	6.8255 sec	0.0256
HSANet	100 (near-perfect)	1.8910 sec	0.01–0.02
MobileNetV3	48.91	25.03 sec	0.49

The accuracy comparison graph illustrates the effectiveness of the extracted features from PCANet and HSANet. PCANet achieves an accuracy of 48.91 percent, indicating limited discriminative capability of its extracted features for 3D palmprint recognition. In contrast, HSANet achieves near-perfect recognition accuracy under the evaluated dataset, demonstrating that attention-guided feature extraction significantly enhances class separability. This substantial improvement confirms that HSANet is more effective in learning robust and distinctive palmprint features compared to PCANet.

ROC Curve Analysis: The ROC curve shows how well each method balances false positives and true positives during recognition. From the results, PCANet performs poorly, with a significantly high Equal Error Rate (EER) of about 0.98. This means that the features extracted by PCANet overlap significantly for genuine and impostor samples, leading to frequent misclassifications. In contrast, HSANet achieves a very low EER close to zero, indicating almost perfect separation between genuine and impostor classes. This indicates that HSANet extracts more reliable and discriminative features, resulting in fewer classification errors and a more dependable biometric system.

CMC Curve Analysis: The CMC curve highlights how accurately each method identifies a person at different ranks. At Rank-1, PCANet achieves a recognition rate of only around

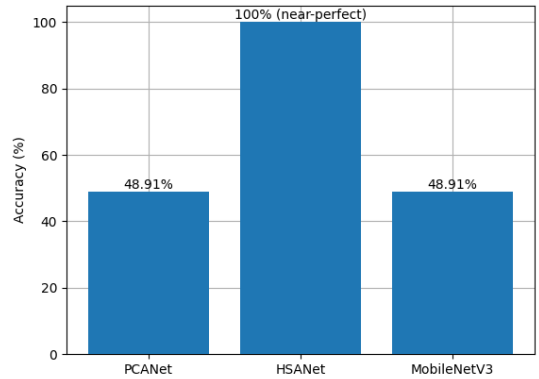

Fig. 6. Accuracy comparison between PCANet,HSANet and MobileNetV3

49 percent, meaning that correct identification often requires searching beyond the first match. On the other hand, HSANet reaches 100 percent Rank-1 recognition, successfully identifying individuals on the first attempt itself. This large performance difference emphasizes the advantage of HSANet's attention-based feature extraction, which allows the system to focus on the most informative palm regions and achieve accurate identification even at the earliest ranking level.

The high recognition accuracy achieved by HSANet is primarily due to controlled acquisition conditions in the IIT Delhi dataset, which exhibits low intra-class variability and minimal noise. This can be attributed to the controlled acquisition conditions, limited dataset variability, and strong discriminative capability of attention-based feature extraction.

On the other hand, PCANet and HSANet enable better class separability by learning decision boundaries and structured feature representations. This illustrates how crucial supervised learning techniques are to obtaining accurate biometric identification.

Although HSANet achieves near-perfect accuracy on the IIT Delhi dataset, reduced intra-class variability and controlled acquisition settings have an impact on this performance. A subject-disjoint training and testing split was used to mitigate potential overfitting, and average performance metrics are reported. However, to verify real-world generalization, more validation on unconstrained datasets is needed.

The superior performance of HSANet can be attributed to its ability to selectively focus on discriminative palm regions such as principal lines, wrinkles, and ridge patterns. Unlike PCANet, which relies on linear feature extraction using PCA filters, HSANet captures complex spatial dependencies through attention mechanisms, resulting in improved feature representation and class separability.

The ROC and CMC curve analysis further validates this observation, where HSANet demonstrates near-ideal behavior with minimal overlap between genuine and impostor distributions. In contrast, PCANet exhibits significant overlap, leading

to higher error rates and reduced identification accuracy.

However, the high recognition accuracy observed on the IIT Delhi dataset should be interpreted with caution. The dataset is collected under controlled acquisition conditions with limited intra-class variability and minimal noise. As a result, the performance may not directly generalize to real-world environments where variations in illumination, pose, and occlusion are more prominent.

MobileNetV3 achieves improved performance compared to PCANet due to its ability to learn hierarchical feature representations using deep convolutional layers. However, its performance remains lower than HSANet, as it does not explicitly incorporate spatial attention mechanisms. While MobileNetV3 provides a good balance between accuracy and computational efficiency, HSANet demonstrates superior performance by focusing on discriminative palm regions through attention-based feature refinement.

The performance of the proposed HSANet-based approach is consistent with recent deep learning-based biometric systems reported in the literature. For instance, multimodal biometric systems combining palmprint and finger knuckle features have demonstrated improved recognition accuracy by leveraging complementary information [1]. Similarly, normalized hand-based biometric frameworks have shown enhanced feature consistency and robustness under controlled conditions [2].

Contactless finger knuckle recognition methods have also achieved promising results; however, they remain sensitive to illumination and pose variations, which can impact reliability in real-world scenarios [4]. Additionally, 3D biometric approaches incorporating depth information have been reported to improve recognition performance, but often rely on specialized acquisition hardware or complex processing pipelines [5], [14].

In contrast to these methods, the proposed framework achieves competitive performance while maintaining computational efficiency and deployability. Unlike existing approaches that require complex hardware setups or deep architectures, the integration of pseudo-3D reconstruction with attention-based feature learning enables robust and discriminative feature extraction using a lightweight and scalable system.

VII. Hardware Results

A. Jetson Nano Board

Designed to execute AI applications at the edge, the NVIDIA Jetson Nano is a compact, low-power embedded AI development board. Instead of depending on cloud processing, it enables machine learning models to operate directly on the device, resulting in quicker and more accurate outcomes. The board's 128-core NVIDIA Maxwell GPU and quad-core ARM Cortex-A57 CPU provide roughly 472 GFLOPS of processing power. The gadget can do tasks including computer vision applications, deep learning, and real-time AI inference thanks to its processing capacity. Additionally, it facilitates the connection of sensors, cameras, and other hardware components needed for AI-based systems by supporting LPDDR4 memory, microSD storage, HDMI, Ethernet, USB ports, and camera interfaces.

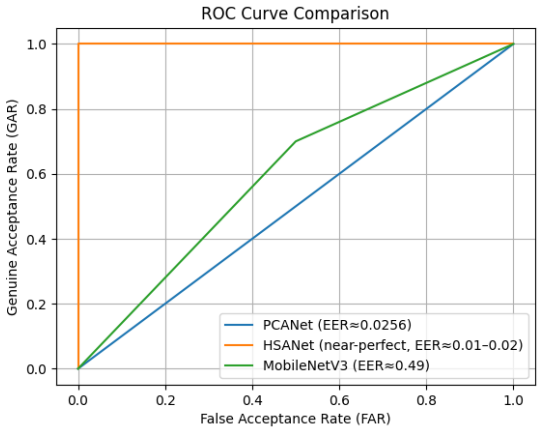

Fig. 7. Shows ROC Curve analysis

The trained HSANet model was installed on the Jetson Nano using an effective inference pipeline. The procedure includes image preprocessing, ROI extraction, HSANet feature extraction, and final classification.

During deployment, the average GPU utilization was observed to be approximately 65–75%, with memory consumption around 2–3 GB. This indicates efficient utilization of available computational resources on the Jetson Nano platform while maintaining stable inference performance.

In order to improve deployment efficiency, model optimization techniques such as reduced precision computing were taken into consideration to speed up inference and reduce memory usage. The system is designed to function within the computational constraints of embedded hardware while guaranteeing consistent recognition performance.

This shows that implementing deep learning-based biometric systems on low-power edge devices is feasible.

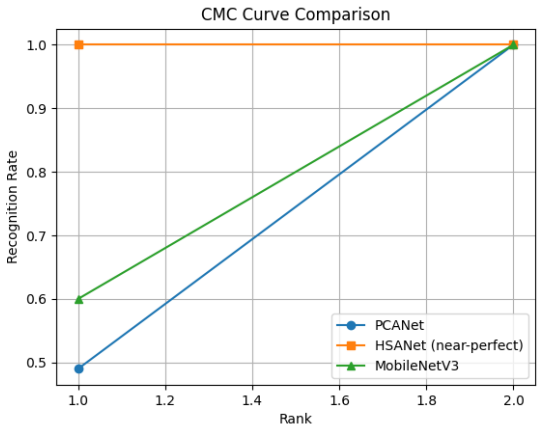

Fig. 8. Shows CMC Curve analysis

Fig. 9. Shows Jetson Orin Nano Super Developer Kit

TABLE II
COMPARISON OF MODEL ACCURACY, COMPUTATION TIME AND EER FOR DIFFERENT ALGORITHMS USING JETSON NANO BOARD

Algorithms	Accuracy in percentage	Computation time	Equal error rate
PCANet	59.12	0.2749sec	0.4185
HSANet	100 (near-perfect)	12.1228sec	0.01–0.02
MobileNetV3	59.31	250.78 sec	40.69

The higher computation time observed for MobileNetV3 on the Jetson Nano is due to the lack of extensive optimization and the limited processing capability of the embedded hardware. Further optimization techniques such as model pruning and quantization can significantly reduce inference time.

The table compares the performance of three algorithms PCANet, HSANet, and MobileNetV3 based on accuracy, computation time, and Equal Error Rate (EER) when implemented on the NVIDIA Jetson Nano board. Among the three

HSANet achieves the highest accuracy of 100 percent (near ideal performance) and an EER of 0.01-0.02,

indicating perfect classification performance, although it requires the highest computation time of 12.1228 seconds. PCANet provides moderate accuracy of 59.12 percent but has the fastest computation time of 0.2749 seconds, making it suitable for faster processing applications. In contrast, MobileNetV3 provides moderate recognition accuracy with improved computational efficiency compared to HSANet. However, its performance is lower than HSANet due to the absence of explicit attention mechanisms for focusing on discriminative palm regions.

Although HSANet achieves near-perfect accuracy on the IIT Delhi dataset, reduced intra-class variability and controlled acquisition settings have an impact on this performance.

The deployment of the proposed system on the Jetson Nano demonstrates its feasibility for real-time biometric authentication in resource-constrained environments. The results indicate that HSANet achieves superior recognition performance even when executed on embedded hardware, validating the robustness of the proposed approach.

However, this improved accuracy comes at the cost of increased computational time compared to PCANet. While PCANet offers faster inference due to its lightweight architecture, its lower recognition accuracy limits its applicability in high-security scenarios. In contrast, HSANet provides a better balance between accuracy and reliability, making it more suitable for applications where precision is critical.

The results also highlight the importance of model optimization techniques for embedded deployment. By reducing model complexity and optimizing inference pipelines, attention-based deep learning models can be effectively adapted for edge AI applications.

Overall, the hardware implementation confirms that the proposed hybrid framework can achieve reliable performance while operating within the computational constraints of low-power devices, making it suitable for practical real-world deployment.

The proposed system is evaluated on a controlled dataset with limited variability. Performance may not generalize to real-world conditions with variations in illumination, pose, and occlusion. Further validation on unconstrained datasets is required.

CONCLUSION

This work presents a hybrid 3D palmprint biometric recognition framework that integrates pseudo-3D reconstruction with attention-based deep learning. The Frankot–Chellappa algorithm is used to generate depth-enhanced representations from single-view 2D images, improving robustness to illumination and positional variations.

A comparative analysis of PCANet, MobileNetV3, and HSANet shows that HSANet achieves superior recognition performance due to its attention-driven feature extraction, while MobileNetV3 provides a balance between accuracy and efficiency.

Experimental results on the IIT Delhi dataset demonstrate high classification accuracy and low error rates. The system is also deployed on the NVIDIA Jetson Nano, confirming its suitability for real-time edge-based biometric authentication.

However, the evaluation is conducted on a controlled dataset with limited variability. Future work will focus on improving robustness and validating the model on unconstrained datasets for real-world deployment.

REFERENCES

[1] D. Bhattacharyya, R. Ranjan, F. Alisherov, and M. Choi, "Biometric authentication: A review," International Journal of u-and e-Service, Science and Technology, vol. 2, pp. 13-28, 2009. Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on July 30,2025 at 08:59:57 UTC from IEEE Xplore. Restrictions apply.

[2] R. Anderson, Security Engineering: A Guide to Building Dependable Distributed Systems, 2nd ed. Hoboken, NJ, USA: John Wiley and Sons, 2008.

[3] (2016). Fingerprint Access Control Systems Market Worth 11.88 Billion By 2024. Available: https://www.grandviewresearch.com/press release/global-fingerprint-access-control-systems-market

[4] N. C. Sickler and S. J. Elliott, "An evaluation of fingerprint image quality across an elderly population vis-a-vis an 18-25 year old population," in Security Technology, 2005. CCST'05. 39th Annual 2005 International Carnahan Conference on, 2005, pp. 68-73.

[5] (2008).The FBI's Next Generation Identification (NGI). Available: http://fingerprint.nist.gov/standard/presentations/archives/NGIOverviewFeb2005.pdf

[6] J. L. Mnookin, "Validity of Latent Fingerprint Identification: Confessions of a Fingerprinting Moderate, The," Law, Prob. and Risk, vol. 7, p. 127, 2008.

[7] A. W.-K. Kong, D. Zhang, and G. Lu, ”A study of identical twins' palmprints for personal verification,” Pattern Recognition, vol. 39, pp. 2149-2156, 2006.

[8] FBI: Palm Print Recognition. Available: https://www.fbi.gov/file repository/about-us-cjis-fingerprints biometrics biometric center of excellences palmprint recognition.pdf/view

[9] Creativity103, “Hand Scan – Interference Collection,” 2016. [Online]. Available: http://creativity103.com/collections/Interference/slides/handscanner.html

[10] V. Kanhangad, A. Kumar, and D. Zhang, ”A unified framework for contactless hand verification,” IEEE transactions on information forensics and security, vol. 6, pp. 1014-1027, 2011.

[11] S. P. Rao, R. Rajendran, S. S. Agaian, and M. M. A. Mulawka, ”Alpha trimmed correlation for touchless finger image mosaicing,” in SPIE Commercial+ Scientific Sensing and Imaging, 2016, pp. 98690U 98690U-12.

[12] S. S. Agaian, M. M. A. Mulawka, R. Rajendran, S. P. Rao, S. K. KM, and S. Rajeev, ”A comparative study of image feature detection and matching algorithms for touchless fingerprint systems,” Electronic Imaging, vol. 2016, pp. 1-9, 2016.

[13] S. Rajeev, S. K. KM, and S. S. Agaian, ”Method for modeling post mortem biometric 3D fingerprints,” in SPIE Commercial+ Scientific Sensing and Imaging, 2016, pp. 98690S-98690S-10.

[14] D. Zhang and G. Lu, 3D Biometrics: Springer, 2013.

[15] W. Zhou, J. Hu, I. Petersen, S. Wang, and M. Bennamoun, ”Performance evaluation of 2D to 3D fingerprint recognition,” in Image and Signal Processing (CISP), 2013 6th International Congress on, 2013, pp. 1736 1741.

[16] Q. Zhao, A. Jain, and G. Abramovich, ”3D to 2D fingerprints: Unrolling and distortion correction,” in Biometrics (IJCB), 2011 International Joint Conference on, 2011, pp. 1-8.

[17] G. Divya, A. S. Mamatha, H. C. Nagaraj, M. N. L, M. R and M. K M, ”Performance Analysis of 3D Finger Knuckle for Biometric Application,” 2025 5th International Conference on Emerging Research in Electronics, Computer Science and Technology (ICERECT), MANDYA, India, 2025, pp. 1-7, doi: 10.1109/ICERECT65215.2025.11378078.

[18] D. Zhang, W.-K. Kong, J. You, and M. Wong, ”Online palmprint identification,” IEEE Transactions on pattern analysis and machine intelligence, vol. 25, pp. 1041-1050, 2003.

[19] A. Kumar, “IIT Delhi Palmprint Database (Version 1.0),” Biometrics Research Laboratory, IIT Delhi. Available: http://www4.comp.polyu.edu.hk/csajaykr/myhome/databaserequest/palm/